TOPIC 1: FORMULAE, EQUATIONS AND AMOUNT OF SUBSTANCE

GRADE 11

SEMESTER 1/2023

Checklist ✓

Students will be assessed on their ability to:

1.1 know the terms 'atom', 'element', 'ion', 'molecule', 'compound', 'empirical formula' and 'molecular formula' ☐

1.2 know that the mole (mol) is the unit for the amount of a substance and be able to perform calculations using the Avogadro constant L (6.02×10^{23} mol^{-1}) ☐

1.3 write balanced full and ionic equations, including state symbols, for chemical reactions ☐

1.4 understand the terms:

- i 'relative atomic mass' based on the 12C scale ☐
- ii 'relative molecular mass' and 'relative formula mass', including calculating these values from relative atomic masses. The term 'relative formula mass' should be used for compounds with giant structures. ☐
- iii 'molar mass' as the mass per mole of a substance in $gmol^{-1}$ ☐
- iv parts per million (ppm), including gases in the atmosphere ☐

1.5 calculate the concentration of a solution in $moldm^{-3}$ and gdm^{-3} ☐

Titration calculations are not required at this stage.

1.6 be able to use experimental data to calculate empirical and molecular formulae ☐

1.7 be able to use chemical equations to calculate reacting masses and vice versa, using the concepts of amount of substance and molar mass ☐

1.8 be able to use chemical equations to calculate volumes of gases and vice versa, using:

- i the concepts of amount of substance ☐
- ii the molar volume of gases ☐
- iii the expression pV = nRT for gases and volatile liquids ☐

1.9 be able to calculate percentage yields and percentage atom economies (by mass) in laboratory and industrial processes, using chemical equations and experimental results

$$Atom\ economy = \frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}}\ x\ 100$$ ☐

1.10 be able to determine a formula or confirm an equation by experiment, including evaluation of the data ☐

1.12 be able to relate ionic and full equations, with state symbols, to observations from simple test-tube experiments, to include:

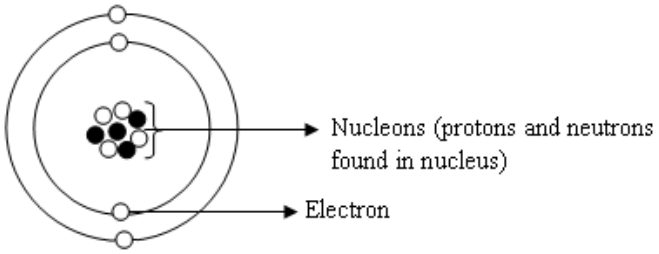
- i displacement reactions ☐
- ii typical reactions of acids ☐
- iii precipitation reactions ☐

1A ATOMS, ELEMENTS AND MOLECULES

SPECIFICATION OBJECTIVE 1.1

Atom

- An atom is the smallest, electrically neutral particle of an element that can take part in a chemical change.
- An atom comprises of subatomic particles **protons**, **neutrons** and **electrons**.
- Protons and neutrons, together known as **nucleons**, are found in the nucleus which is the center of the atom.
- Electrons orbit around the nucleus in definite energy levels called **shells**.
- An atom gets its chemical properties from the number of electrons and arrangement of electrons in the atom.



- **The masses and charges of the three sub atomic particles**

Particle	Symbol	Charge/C	Relative charge	Mass/g	Relative mass
Proton	p	$+1.60 \times 10^{-19}$	+1	1.67×10^{-24}	1
Neutron	n	0	0	1.67×10^{-24}	1
Electron	e^-	-1.60×10^{-19}	-1	9.11×10^{-28}	1 / 1840

- The mass of electrons are negligible, hence the mass of the atoms are determined by the mass of nucleons.
- Atoms in an element are represented by using the following notation.

Mass number = sum of protons and neutron in the nucleus of an atom

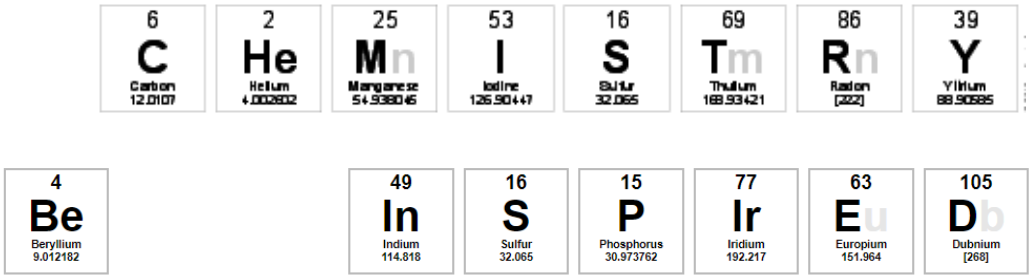
Atomic number = protons in the nucleus of an atom

$$^{A}_{Z}X$$

- The number of protons is equal to the number of electrons; thus, an atom is electrically neutral.

Element

- Elements are made up of only one type of atoms.
- Elements are chemically the simplest substance, so they cannot be broken down using physical/ chemical means.
- The atoms of an element all contain the same number of protons.
- Can exist as either atoms (eg: sodium, argon) or molecules (eg: oxygen (O_2), nitrogen (N_2)).
- Each element has its own characteristic properties.
- Chemists use symbols to represent elements.



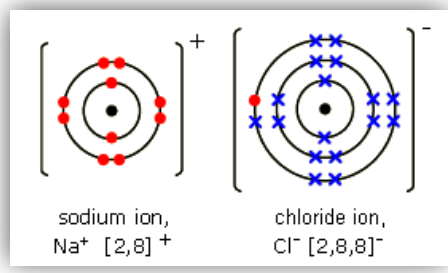
Molecule

- A molecule is the smallest, electrically neutral particle, of an element or compound that can exist on its own.
  Example: Oxygen molecules (O_2) - element
  Water (H_2O) – compound
- It is also the name given to the smallest part of those elements which do not exist as atoms in the free state i.e. hydrogen H_2, oxygen O_2, nitrogen N_2, fluorine F_2, chlorine, Cl_2, bromine Br_2 and iodine I_2.
- A molecule consists of two or more atoms of the same element or different elements, chemically combined.
- Ionic compounds (e.g. sodium chloride) do not exist as molecules.

Compound

- A compound is a substance formed when two or more elements chemically combined. Example: water, sugar, sodium chloride
- Have properties that are different from its component elements.
- Can be broken down into simpler substances by chemical methods.
- Chemical formulas are used to show the composition of compounds.
  Example: NaCl, H_2O.

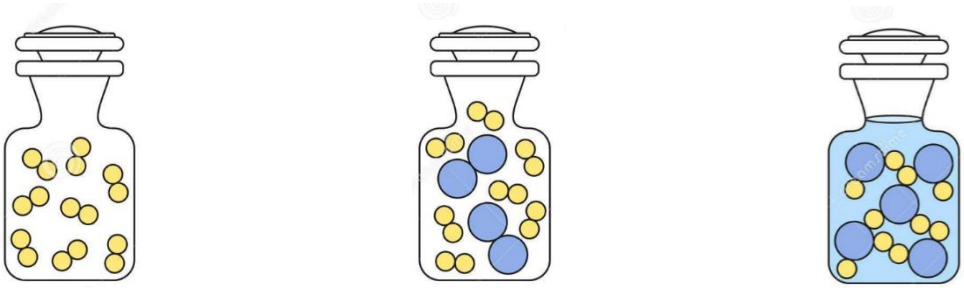
Ion



- An ion is an atom or a group of atoms (molecule) which carries an electrical charge.
- Negative ion has more electron than protons. Example: Cl^- ion has 17 protons and 18 electrons.
- A negative ion is obtained by adding electrons to a neutral atom.
- A positive ion has fewer electrons than protons. Example: Na^+ ion has 11 protons and 10 electrons.
- A positive ion is obtained by removing electrons from a neutral atom.

EXERCISE 1

Classify the following into elements, compounds, mixtures and molecules



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CHECKPOINT 1

1. Classify each of these symbols and formulae as atoms, molecules or ions.

Ne CO_2 H^+ S_8 Al^{3+}

2. Which of these formulae represent elements, compounds, or neither elements nor compounds? Explain your answer.

Br_2 H_2O_2 NO_3 O_3 CaO

1B WRITING CHEMICAL EQUATIONS

SPECIFICATION OBJECTIVE 1.3

Chemical equations

- Shows the formulae of the reactants and the products.
- Shows the physical state in which it exists.
- Shows the relationship between the numbers of each substance involved; this is known as the **stoichiometry** of the reaction.

- Equations can be written in words or using chemical formula.

Example: hydrogen + oxygen → water

$H_2 + O_2 \rightarrow H_2O$

- Chemical equations could be either **molecular equations** or **ionic equations**.

Balancing molecular equations

- A balanced chemical equation must have the same number of each kind of atoms on the left-hand side (LHS) of the equation as on the right-hand side (RHS).
- Net charge on both reactants and products side should be 0.
- Simplest whole number ratios should be used to balance the equations.
- Steps to follow in writing chemical equations

1. Work out what has reacted and what has been formed. Write a word equation.
2. Get the correct formula for each species. Include the state symbols where necessary. When the correct formula of a species has been obtained, **do not change the formula** to balance the equation.
3. Having written out the initial equation, check to see if it is balanced. An equation balances if the **same number of each type of atom appears on either side of the arrow**.
4. Place whole numbers in front of any formula to indicate if more than one of it is required. This multiplies everything in the formula immediately behind it. Continue until the equation balances.
5. Finally, check the equation to see that you have balanced it correctly.

SO_4^{2-}

SO_3^{2-}

NO_3^-

NO_2^-

- ide - single atom ion
- ate - polyatomic ion
- ite - one O less, ate

Worked example	
Step 1	sodium + water → sodium hydroxide + hydrogen
Step 2	$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$
Step 3	Count up the atoms : LHS ... 1 x Na, 2 x H, 1 x O RHS ... 1 x Na, 3 x H, 1 x O.
	The equation doesn't balance; an extra H is needed on the LHS. However formulae should not be changed. One can only get extra H's by having two waters; multiply H_2O by two.
Step 4	$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$
	Now there are too many O's (2) and H's (4) on the LHS; multiplying the NaOH by two will solve this problem.
	$Na_{(s)} + 2\ H_2O_{(l)} \rightarrow 2\ NaOH_{(aq)} + H_{2(g)}$
	However, it creates yet another problem because it has introduced an extra Na on the RHS; multiply the Na on the LHS by two.
	$2\ Na_{(s)} + 2\ H_2O_{(l)} \rightarrow 2\ NaOH_{(aq)} + H_{2(g)}$
Step 5	Check the equation to check whether it is balanced. LHS ... 2 x Na, 4 x H, 2 x O RHS ... 2 x Na, 4 x H, 2 x O. It is balanced.

Balancing ionic equations

- Many reactions involve ions in solution.
- When substances react in this way an ionic equation is useful.
- In ionic equations, only the reacting particles are included.

Worked example	
Write down a balanced ionic equation for the following reaction. $CaCO_3\ (s) + HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$	
Step 1	Balance the molecular equation first $CaCO_3\ (s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$
Step 2	Separate the ions in solution (write full molecular formula for solids, liquids and gases) and cancel out the **spectator ions** $CaCO_3\ (s) + 2H^+\ (aq) + \cancel{2Cl^-(aq)} \rightarrow Ca^{2+}\ (aq) + \cancel{2Cl^-(aq)} + CO_2(g) + H_2O(l)$
Step 3	Then write down the final ionic equation with state symbols $CaCO_3\ (s) + 2H^+\ (aq) \rightarrow Ca^{2+}(aq) + CO_2(g) + H_2O(l)$

EXERCISE 2

Question 1: Balance the following equations

a) $Ca(OH)_2 + HNO_3 \rightarrow Ca(NO_3)_2 + H_2O$

b) $Mg(NO_3)_2 \rightarrow MgO + NO_2 + O_2$

c) $Cu + H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$

d) $Al + O_2 \rightarrow Al_2O_3$

e) $KOH + Cl_2 \rightarrow KCl + KClO_3 + H_2O$

CHECKPOINT 2

1. Sodium thiosulfate ($Na_2S_2O_3$) solution reacts with dilute hydrochloric acid to form a precipitate of sulfur, gaseous sulfur dioxide and a solution of sodium chloride. Write an equation, including state symbols, for this reaction.

2. Solutions of ammonium sulfate and sodium hydroxide are warmed together to form sodium sulfate solution, water and ammonium gas. Write the simplest ionic equation for this reaction.

3. In some airbags, solid sodium azide (NaN_3) decomposes forming nitrogen gas and sodium as the only products. Write an equation for the decomposition of sodium azide. State symbols are not required.

1B TYPICAL REACTIONS OF ACIDS

SPECIFICATION OBJECTIVE 1.12(ii)

Acids with metals

General equation; Acid + Metal → Salt + Hydrogen

Acids with metal oxides/metal hydroxides

General equation; Acid + Metal oxide/hydroxide → Salt + Water

Acids with carbonates

General equation; Acid + Metal carbonate → Salt + Water + carbon dioxide

EXERCISE 3

Write a balanced chemical equation for the following reactions

a. Aluminium with sulfuric acid

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b. Copper(ll)oxide with hydrochloric acid

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c. Sodium hydroxide with phosphoric acid

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d. Lithium carbonate with nitric acid

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CHECKPOINT 3

1. Write full ionic equations for the reaction between;
   a. Zinc and sulfuric acid
   b. Aluminium oxide and hydrochloric acid.
2. Write the simplest ionic equations for the reactions between;
   a. Zinc and hydrochloric acid
   b. Magnesium carbonate and nitric acid.

1B DISPLACEMENT REACTIONS

SPECIFICATION OBJECTIVE 1.12(i)

- In simple, it is a reaction in which one element replaces another element in a compound. Example:

$$Mg(s) + CuSO_4(aq) \rightarrow Cu(s) + MgSO_4(aq)$$

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + Al_2O_3(s)$$

Remember a more reactive metal displaces the less reactive metal from its compound.

1B PRECIPITATION REACTIONS

SPECIFICATION OBJECTIVE 1.12(iii)

- A precipitation reaction refers to the formation of an insoluble salt when two solutions containing soluble salts are combined.
- The insoluble salt that falls out of solution is known as the precipitate.

Example:

$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

$NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

$Ca(OH)_2(aq) + CO_2 \rightarrow CaCO_3(s) + H_2O(l)$

SOLUBILITY RULES

SOLUBLE	INSOLUBLE
All nitrates	None
Most sulfates	Lead sulfate, barium sulfate and calcium sulfate
Most chlorides, bromides and iodides	Silver chloride, silver bromide, silver iodide, lead chloride, lead bromide, lead iodide
Sodium carbonate, potassium carbonate, ammonium carbonate	Most other carbonates
Sodium hydroxide, potassium hydroxide, ammonium hydroxide	Most other hydroxides

EXERCISE 4 .....

Complete the following equations

a. $Cl_2(aq) + NaBr(aq) \rightarrow$ .....

b. $Cu_{(s)} + AgNO_{3(aq)} \rightarrow$ ....

c. $(NH_4)_2SO_4(aq) + BaCl_2(aq) \rightarrow$ .....

d. $Na_2S(aq) + NiSO_4(aq) \rightarrow$ .....

e. $Al(NO_3)_3(aq) + Na_3PO_4(aq) \rightarrow$ .....

1C COMPARING MASSES OF SUBSTANCES

SPECIFICATION OBJECTIVE 1.1, 1.2 & 1.4

Relative masses of elements and compounds

- Measuring the mass of an atom or compound using a balance would be very difficult as they are very small.
- Hence, to measure the mass of an atom or a compound the ***atomic mass unit*** (amu) is used as a standard reference.
- The relative atomic mass does not have any units. They are measured compare to amu.
    - ***Atomic mass unit*** is the mass of a carbon-12 atom divided by 12. This is used as the standard reference scale to measure mass of atoms and molecules.
    - ***Relative atomic mass*** of an element is the weighted average mass of an atom of that element divided by 1/12th the mass of a carbon-12 atom.

$$Ar = \frac{mean\ mass\ of\ an\ atom\ of\ an\ element}{\frac{1}{12} of\ the\ mass\ of\ an\ atom\ of\ \ ^{12}C}$$

    - ***Relative molecular mass*** of a molecule or compound is the sum of the relative atomic masses of all the atoms in the chemical formula.
    - ***Relative isotopic mass*** is the mass of an atom of that isotope divided by 1/12th the mass of a carbon-12 atom.

Mole: amount of substance

- When chemists measure how much of a particular chemical reacts they measure the amount in grams or the volume of a gas. However, chemists find it convenient to use a unit called **mole**.
- The **mole** is the amount of substance which contains the same number of particles (atoms, ions, molecules, formulae or electrons) as there are carbon atoms in 12 g of carbon -12.
- The number of particles in a mole is known as **Avogadro's constant** which is equal to **6.02 x 10^{23} mol^{-1}**.

1 mole = 6.02 x 10^{23}particles *(it can be atoms, ions, molecules)*

Example:

6.02 X 10^{23} helium atoms in 4g of He

6.02 X 10^{23} carbon dioxide molecules in 44g of CO_2

6.02 X 10^{23} nitrate ions in 62 g of NO_3^-

WORKED EXAMPLE
How many water molecules are there in 1.25g of water? • Moles of water = 1.25/18 = 0.0694 • I mol of water → 6.02×10^{23} moleclues 0.0694 mols → *x* *X* = 4.18×10^{22}

- **The molar mass** of a substance is the mass, in grams, of one mole.

Number of moles = mass in grams / molar mass of

- **The molar volume** of a gas is the volume occupied by one mole at room temperature and pressure. It is equal to 24 dm^3 at room temperature and pressure (r.t.p).

Number of moles of gas = volume of gas (dm^3) / molar volume (24 dm^3)

EXERCISE 5

Question 1: Calculate the number of atoms in 1 mole of O_2

1.204×10^{24} atoms

Question 2: Calculate the number of atoms in 0.5 mole of SO_2.

9.03×10^{23} atoms

Question 3: Calculate the number of ions in 0.5 mole of $Ba(OH)_2$ solution.

9.03×10^{23} ions

Question 4: Calculate the number of moles of oxygen atoms in 64 g of oxygen atoms.

4 moles

Question 5: Calculate the number of moles of oxygen molecules in 4g. Oxygen molecules have the formula O_2.

0·125mol

Question 6: Calculate the mass of 3 moles of sulfur dioxide SO_2.

193·3g

Question 7: Calculate the number of moles of hydrogen molecules in 240 cm^3 of the gas.

0·01mol

Question 8: Calculate the volume of 0.005 moles of carbon dioxide at r.t.p.

120 cm^3

1C CALCULATIONS USING REACTING MASSES

SPECIFICATION OBJECTIVE 1.7

WORKED EXAMPLE 1

The equation for the reaction is

$$SO_3 + H_2O \rightarrow H_2SO_4$$

What mass of sulfur trioxide is needed to form 75g of sulfuric acid?

$$\frac{mass}{R \times Mr}$$

$$\frac{SO_3}{\frac{x}{80.1 \times 1}} : \frac{H_2SO_4}{\frac{75g}{98.1 \times 1}}$$

$$98.1x = 75 \times 80.1$$

$$\frac{98.1x}{98.1} = \frac{6007.5}{98.1}$$

$$x = 61.249$$

WORKED EXAMPLE 2

The equation for the reaction is

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

What mass of ammonia is needed to form 100g of ammonium sulfate?

25.749

WORKED EXAMPLE 3

A 16.7g of sodium carbonate ($Na_2CO_3.10H_2O$) is heated at a constant temperature for a specified time until the reaction is complete. A mass of 3.15g of water is obtained. What is the equation for the reaction occurring?

$Mr = 286$

$$Na_2CO_3 \cdot \underbrace{10H_2O}_{\text{water of crystallisation}}{}_{(s)} \xrightarrow{\Delta} Na_2CO_3 \cdot 7H_2O_{(s)} + 3H_2O_{(g)}$$

16.7g 3.15g

$$\begin{bmatrix} Na - 23 \\ C - 12 \\ O - 16 \\ H - 1 \end{bmatrix}$$

$mol = \frac{mass}{Mr}$

$Na_2CO_3 \cdot 10H_2O$	:	H_2O
$= \frac{16.7g}{286}$		$\frac{3.15g}{18}$
$= 0.05839$		0.175
$\frac{0.05839}{0.05839}$		$\frac{0.05839}{0.05839}$
1	:	3

WORKED EXAMPLE 4

An oxide of copper is heated in a stream of hydrogen to constant mass. The masses of copper and water formed are 17.6g and 2.56g respectively. Write the equation for the reaction.

Cu^{+} | Cu^{2+}

$$CuO_2 + H_2 \longrightarrow 2\underbrace{Cu}_{17.6g} + \underbrace{H_2O}_{2.56g}$$

$mol = \frac{mass}{Mr}$

Cu	:	H_2O
$\frac{17.6}{63.5}$		$\frac{2.56}{18}$
$\frac{0.2772}{0.14222}$		$\frac{0.14222}{0.14222}$
2	:	1

EXERCISE 6

Question 1: What mass of lead (II) sulfate would be produced by the action of excess dilute sulfuric acid on 10 g of lead nitrate dissolved in water?

$$Pb(NO_3)_{2(aq)} + H_2SO_{4(aq)} \rightarrow PbSO_{4(s)} + 2HNO_{3(aq)}$$

Pb - 207.2
N - 14 × 2
O - 16 × 6 +

$M_r = 331.2$

$M_r = 303.3$

① mol $Pb(NO_3)_2$

$mol = \frac{mass}{M_r}$

$= \frac{10g}{331.2}$

$= 0.03019$

② mol $PbSO_4$ (ratio)

1 : 1

$= 0.03019$

③ final (mass $PbSO_4$)

mass = mol × M_r

$= 0.03019 \times 303.3$

$= 9.16g$

2.d.p / 3.s.f

$= 9.16g$

Question 2: What is the total volume of gas produced by the action of heat on 1 g of silver nitrate?

$$2AgNO_{3(s)} \rightarrow 2Ag_{(s)} + 2NO_{2(g)} + O_{2(g)}$$

① Mol known
② mol Unknow (Ratio)
③ final answer.

$0.212 dm^3$

Question 3: When excess carbon dioxide is passed into sodium hydroxide solution, sodium carbonate solution is formed. This can be crystallized out as $Na_2CO_3.10H_2O$. What mass of crystals would be produced from 5 g of sodium hydroxide in excess water? (Note: *You need the water expressed as moles in the equation)*

$$2NaOH_{(aq)} + CO_{2\,(g)} + 9H_2O_{(l)} \rightarrow Na_2CO_{3\,(aq)} + 10H_2O_{(l)} \rightarrow Na_2CO_3.10H_2O_{(s)}$$

17.875g

Question 4: What volume of sulfur trioxide would be produced by the complete reaction of 100 cm^3 of sulfur dioxide with oxygen? What volume of oxygen would be needed to just react with the sulfur dioxide?

50cm^3

Question 5: What volume of ammonia would be produced if 10 cm^3 of nitrogen was reacted with 20cm^3 of hydrogen?

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

(Note: The reacting volumes given in the question are not the same as those in the reaction. You must have an excess of one of the gases.)

13.33 cm^3

Question 6: 13.2 g of a sample of zinc sulphate, $ZnSO_4.xH_2O$, was strongly heated until no further change in mass was recorded. On heating, all the water of crystallisation evaporated as follows:

$$ZnSO_4.xH_2O \rightarrow ZnSO_4 + xH_2O.$$

Calculate the number of moles of water of crystallisation in the zinc sulphate sample given that 7.4 g of solid remained after strong heating.

1 : 7

1C YIELD OF A REACTION

SPECIFICATION OBJECTIVE 1.9

Percentage yield

The efficiency of a reaction can be measured by the percentage yield. This gives the amount of product loss and how wasteful the process is.

- The percentage yield of a reaction can be calculated using the formula;

$$\text{Percentage yield} = \frac{\text{Actual (experimental) yield}}{\text{Theoretical yield}} \times 100$$

- **Theoretical yield** - the mass of the product that should be formed. It can be calculated using the mass of reactants and balanced chemical equation.
- For any reaction the **actual mass of product formed** (actual yield) **will always be less than the theoretical yield.**
- This is because
  - all the reactants do not undergo complete reaction;
  - some product is lost during handling,
  - solution loss during filtration or during transfer between containers.

WORKED EXAMPLE 1
Copper(II)carbonate is decomposed to obtain copper(II)oxide. The equation for the reaction is $CuCO_3 \rightarrow CuO + CO_2$ What is the theoretical yield of copper(II)oxide obtainable from 5.78g of Copper(II)carbonate? 3.72g

WORKED EXAMPLE 2

Magnesium phosphate can be prepared from magnesium by reacting it with phosphoric acid. The equation for the reaction:

$$3Mg + 2H_3PO_4 \rightarrow Mg_3(PO_4)_2 + 3H_2$$

What is the theoretical yield of magnesium phosphate obtainable from 5.62g of magnesium?

20.27g

WORKED EXAMPLE 3

A manufacturer uses this reaction to obtain methanol from carbon monoxide and hydrogen:

$$CO + 2H_2 \rightarrow CH_3OH$$

The manufacturer obtains 4.07 tonnes of methanol starting from 4.32 tonnes of carbon monoxide. What is the percentage yield?

82.4 %

Atom economy

- It is also a method used to measure the efficiency of a reaction.
- The atom economy of a chemical reaction is a measure of the proportion of reactant atoms that become part of the desired products rather than by-products.
- A reaction that has a very high yield could still be very wasteful if a lot of the atoms from the reactants end up in by-products.
- **Inefficient, wasteful processes** have **low atom economies**.
- **Efficient processes** have **high atom economies**, and are important for sustainable development, as they use fewer natural resources and create less waste.
- The atom economy of a reaction can be calculated:

$$\% \text{ Atom economy} = \frac{\text{Molecular mass of the desired product}}{\text{Sum of the molecular masses of all products}} \times 100$$

- **Addition reactions** have 100% atom economy since only one product will be formed and no atoms are wasted as by-products.
- **Substitution reactions** -have a lower atom economy than addition reactions. These reactions would form at least two products, where only one would be the desired product.
- Companies in the chemical industry will often choose reactions with higher atom economies. Keeping high atom economy (greener chemical processes) has environmental and economic benefits.
- Low atom economy leads to more waste. This makes separation of desired product and disposal of waste products costly.
- Also the process itself becomes more costly and less efficient if most of the reactants end up as waste as more raw materials would be required to get sufficient amount of product.

WORKED EXAMPLE 1

What is the atom economy for making hydrogen by reacting coal with steam?

$$C(s) + 2H_2O(g) \rightarrow CO_2(g) + 2H_2(g)$$

C – 12, H – 1, O – 16

$$\% AE = \frac{4}{44+4}$$

$$= 8.33\%$$

WORKED EXAMPLE 2
Hydrazine can be used as a rocket fuel and is manufactured using the reaction: $2NH_3 + NaOCl \rightarrow N_2H_4 + NaCl + H_2O$ What is the atom economy for this reaction? 29.5%

EXERCISE 7

Question 1: 25g of Fe_2O_3 was reacted with carbon monoxide and it produced 10g of Fe. What is the percentage yield?

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Question 2: Sodium carbonate is an important industrial chemical manufactured by the Solvay process. The overall equation for the process is:

$$CaCO_3 + 2NaCl \rightarrow Na_2CO_3 + CaCl_2$$

A manufacturer starts with 75kg of calcium carbonate and obtains 76.5kg of sodium carbonate. Calculate the percentage yield and atom economy for this reaction.

17.5g | 57.1 %

Question 3: A manufacturer of ethene wants to convert some ethane into 1,2-dichloroethane. He considers two possible reactions:

Reaction 1 $H_2C{=}CH_2 + Cl_2 \rightarrow ClCH_2CH_2Cl$

Reaction 2 $2H_2C{=}CH_2 + 4HCl + O_2 \rightarrow 2ClCH_2CH_2Cl + 2H_2O$

Explain, without doing calculation, which reaction would be a good choice on the basis of atom economy.

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1D EMPIRICAL AND MOLECULAR FORMULA

SPECIFICATION OBJECTIVE 1.1, 1.6 & 1.8(iii)

- **Empirical formula** gives the smallest whole number ratio of atoms in a compound.
- **Molecular formula** gives the actual number of atoms in a molecule.
- Sometimes, empirical and molecular formula are the same.

WORKED EXAMPLE 1

Calculate the empirical formula of a compound with C (1.8g), O (0.48g), H (0.3g)

	C	H	O
1) Write out ratios by mass	1.8	0.3	0.48
2) Divide by relative atomic mass	1.8 / 12	0.3 / 1	0.48 / 16
(this gives the molar ratio)	0.15	0.3	0.03
3) If not whole numbers then scale up			
- try dividing by smallest value (0.03)	5	10	1
4) Express as a formula	$C_5H_{10}O$		

WORKED EXAMPLE 2

Calculate the molecular formula of a compound of empirical formula CH_2 and relative molecular mass 84.

1) Calculate empirical formula mass (CH_2 unit) = 14
2) Divide molecular mass (84) by empirical formula mass = 84/14 = 6
3) molecular formula = empirical formula (CH_2) x 6 = C_6H_{12}

EXERCISE 8

Question 1: Find the empirical formula of the compound containing 48.70% C, 8.10% H and 43.20% O by mass.

Question 2: A sample of a compound has the composition: sodium 9.20g; sulfur 12.8g; oxygen 9.60g. Find the empirical formula.

Question 3: An organic compound has the following percentage composition by mass: 38.7% carbon; 51.6% oxygen; 9.7% hydrogen. Find its empirical formula.

Question 4: A drum of white powder, compound X, has been dumped in the river and may cause a pollution problem. The laboratory analysis of a sample of X shows that it contains 2.66g of calcium, 0.80g of carbon and 3.22g of oxygen. Calculate the empirical formula of X to decide whether this compound is a hazard.

① $mol = \frac{mass}{Mr}$

② Smallest Ratio (divide by smallest)

Ca	:	C	:	O
$\frac{2.66}{40.1}$		$\frac{0.8}{12}$		$\frac{3.22}{16}$
$\frac{0.06633}{0.06633}$	:	$\frac{0.06667}{0.06633}$	:	$\frac{0.2013}{0.06633}$
1	:	1	:	3

$CaCO_3$

Question 5: Glucose has the empirical formula CH_2O and relative molecular mass 180. Find the molecular formula of glucose.

$(CH_2O)n$

$n = \frac{Mr}{Emp.Mr} = \frac{180}{30} = 6$

$(CH_2O)6$

$C_6H_{12}O_6$

Question 6: 4.6 g of an alcohol, with molar mass 46 $gmol^{-1}$, is burnt in excess oxygen. It produces 8.8 g of carbon dioxide and 5.4g of water. What is the empirical formula and the molecular formula of the alcohol?

Ideal gas equation pV =nRT

- pV = nRT is the ideal gas equation and can be used for gases (or volatile liquids above their boiling points) to find the amount of substance in moles.
- If the mass of the substance is known, then the molar mass of the compound can be calculated.
- This gives extra information needed to workout the empirical and molecular formula of the compound
- SI units used in the ideal gas calculations are:
    - p = pressure in pascals (Pa)
    - V = volume in cubic meters (m^3)
    - T = temperature in kelvin (K)
    - n = amount of substance in moles (mol)
    - R = the gas constant ($8.31 Jmol^{-1}K^{-1}$)

Conversion tips when doing questions

CONVERSION	HOW TO DO IT
kPa → Pa	Multiply by 10^3
cm^3 → m^3	Divide by 10^6
dm^3 →m^3	Divide by 10^3
°C → K	Add 273

WORKED EXAMPLE 1
A 0.280g sample of a gas has a volume of 58.5 cm^3, measured at a pressure of 120kPa and temperature of 70°C. calculate the molar mass of the gas.

WORKED EXAMPLE 2
A compound has the percentage composition by mass C = 52.2%, H = 13.0%, O = 34.8%. A sample containing 0.173g of the compound had a volume of 95.0cm^3 when measured at 105kPa and 45°C. What is the molecular formula of the compound?

EXERCISE 9

Question 1: A 2.82g of sample of gas has a volume of 1.26dm^3, measured at a pressure of 103kPa and temperature of 55°C. Calculate the molar mass of the gas

① mol $n = \frac{PV}{RT}$

$V = 1.26 \times 10^{-3} m^3$

$P = 103 \times 10^3 pa$

$T = 55°C + 273 K$

$n = \frac{103 \times 10^3 \times 1.26 \times 10^{-3}}{8.31 \times 328}$

$= \frac{129.78}{2725.68}$

$= 0.047614 mol$

② $M_r = \frac{mass}{mol}$

$= \frac{2.82}{0.047614}$

$= 59.23$

$= 59.29$

Question 2: A compound has the composition by mass C = 40.0%, H = 6.7%, O = 53.3%. A sample containing 0.146g of the compound had the volume of 69.5 cm^3 when measured at 98kPa and 63°C. What is the molecular formula of the compound?

① mol of sample $n = \frac{PV}{RT}$

- Conversions

$V = 69.5 \div 10^6\ m^3$ [2]

$P = 98 \times 10^3\ Pa$

$T = 63°C + 273\ K$

mol (n) = 2.43×10^{-3} mol [1]

② Mr of sample

$Mr = \frac{mass}{mol} = \frac{0.146}{2.43 \times 10^{-3}}$

$= 59.8$

$= 60\ g\,mol^{-1}$ [1]

③ Empirical formula

C	:	H	:	O
$\frac{40}{12}$		$\frac{6.7}{1}$		$\frac{53.3}{16}$
$\frac{3.33}{3.33}$		$\frac{6.7}{3.33}$		$\frac{3.33}{3.33}$
1	:	2	:	1

CH_2O [1]

④ Molecular formula (n)

$n = \frac{Mr}{Emp.Mr} = \frac{60}{30}$

$n = 2$

M.f = $C_2H_4O_2$ [1]

1E MOLAR VOLUME CALCULATIONS

SPECIFICATION OBJECTIVE 1.8(ii), 1.4(iv) &

Calculations involving chemicals in solutions

- Many chemical reactions occur in solution.
- Concentration of a solution is measured in moles per cubic decimeter (mol dm^{-3})

1 litre = 1 dm^3 = 1000cm^3

Example: Sodium chloride can be labeled as 1M NaCl. This means that each dm^3 (1000 cm^3) of the solution contains 1 mole (58.5 g) of NaCl, or its concentration is 1 mol dm^{-3}.

[*It doesn't mean that 58.5g of NaCl have been added to 1000 cm^3 of water as the volume of the mixture may no longer be 1000 cm^3. The solution is made by dissolving 58.5g of the solid in about 500cm^3 and then adding more water to make it up to 1000cm^3*]

- Concentration in moldm^{-3} is called **molarity**.

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{Volume of solution (dm}^3\text{)}}$$

- To convert molarity to a unit of gdm^{-3} ,

$$\text{Concentration in gdm}^{-3} = \text{Molarity x Molar mass of the solute}$$

EXERCISE 10

Question 1: In an experiment you need a 0.150M solution of sodium chloride. How many grams of sodium chloride would you need to make 500cm^3 of 0.150M sodium chloride solution?

Question 2: If you weigh out 10.60g of sodium carbonate, Na_2CO_3 and make up to 250 cm^3 in a volumetric flask, what will be the molarity of the resulting solution?

Question 3: 25 cm^3 of 0.10 mol dm^{-3} NaOH react with 50 cm^3 of a solution of H_2SO_4.What is the molarity of the H_2SO_4? What is its concentration in gdm^{-3}?

$$2NaOH_{(aq)} + H_2SO_{4\,(aq)} \rightarrow Na_2SO_{4\,(aq)} + 2H_2O_{(l)}$$

Concentrations in PPM (parts per million)

- This unit is used when describing **very low concentrations**.
- Percent means out of hundred, Parts per million means out of a million
- It usually describes something dissolved in water.
- Parts per million (ppm) is often used for levels of pollutants in air or water.
- One ppm is equivalent to one milligram of a solute per liter of water (mg/l) or one milligram of something per kg of solvent (mg/kg).

***Note**: Parts per billion and parts per trillion are also widely used when measuring polluting chemicals present in very low concentrations.*

$$\text{Concentration in ppm} = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 1000\,000$$

SOLUTIONS IN PPM

WORKED EXAMPLE 1
A solution contains 0.176g of solute dissolved in 750g of solvent. What is the concentration in ppm?

$$ppm = \frac{0.176\,g}{750\,g} \times 10^6$$

$$= 234.67\ ppm$$

Meaning there will be 234.67g of solute dissolved in 1000,000g of solvent.

WORKED EXAMPLE 2

A mass of 23mg of sodium chloride is dissolved in 900g of water. What is the concentration of sodium chloride in the solution in ppm?

$\frac{23mg}{1000} \rightarrow 0.023g$

$ppm = \frac{0.023g}{900g} \times 10^6$

$= 25.56 \text{ ppm}$

WORKED EXAMPLE 3

A sample of river water contains phosphate ions with a concentration of 17ppm. What is the mass of the phosphate ions in 500g of water?

$ppm = \frac{\text{mass of S}}{\text{mass solv}} \times 10^6$

$\frac{17}{1} = \frac{x}{500g} \times 10^6$

$\frac{17 \times 500}{10^6} = \frac{x \times 10^6}{10^6}$ $x = 8.5 \times 10^{-3} g$ $0.0085g$

Concentration in ppm = $\frac{\text{volume of gas}}{\text{Volume of air}}$ x 1000 000

GASES IN PPM

WORKED EXAMPLE 4

Some nitrogen dioxide gas, with a volume of 1.5dm^3, mixes with 10000dm^3 of air. What is the concentration of nitrogen dioxide, in ppm, in air?

$ppm = \frac{1.5dm^3}{10,000dm^3} \times 10^6$

$= 150 \text{ ppm}$

WORKED EXAMPLE 5

5000dm^3 of air is found to contain ozone with a concentration of 87ppm. What volume of ozone is in the sample of air?

$$ppm = \frac{\text{vol of sol}}{\text{vol of soln}} \times 10^6$$

$$\frac{87}{1} = \frac{x}{5000\ dm^3} \times 10^6$$

$$= 0.435\ dm^3$$

WORKED EXAMPLE 6

Two samples of air containing sulfur dioxide was analysed. The results for sample 1 showed that 500dm^3 of air contained 37cm^3 of sulfur dioxide. The results of sample 2 showed that there were 1.4dm^3 of sulfur dioxide in 4000dm^3 of air. Show, by calculation, which sample has the higher concentration, in ppm, of sulfur dioxide.

EXERCISE 11

Question 1: There is 2 g of organophosphate pollution in 1.5 million grams of river water. Calculate the concentration of organophosphate in parts per million.

Question 2: Suppose 17 grams of sucrose is dissolved in 183 grams of water. What is the concentration of sucrose in ppm?

Question 3: A certain pesticide has a toxic solubility of 5.0 grams/Kg of body weight. What is this solubility in ppm?

Question 4: 200dm^3 of air contains 58cm^3 of chlorine. What is the concentration of chlorine in ppm?